A FLAT MORPHISM SENDS CODIMENSION ZERO POINTS TO CODIMENSION ZERO POINTS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-morphisms.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The statement is displayed as an implication with an arrow-shaped sentence; the content is the assertion that codimension zero is preserved by a flat morphism.

Editorial note. The two references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: the three native excerpt spans of sources/stacks-spaces-morphisms/preamble.tex (lines 8-13, 14-18 and 154-217). The chapter file itself carries no preamble text.

Allowed standard prerequisites (editorial).

- Codimension zero points characterized by etale neighbourhoods. Properties of Spaces, Lemma spaces-properties-lemma-codimension-0-points, used at proof lines 10316-10317. Reduces the claim to generic points of irreducible components.
- Generalizations lift along flat morphisms of schemes. Morphisms, Lemma morphisms-lemma-generalizations-lift-flat, used at proof lines 10319-10321. Is the scheme statement that settles the translated claim.
- Etale localization of the statement. used at proof lines 10316-10318. Allows the reduction to schemes.

Source reference. Original lemma 03HV: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4581–4581. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 0BAQ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 14218–14218. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let S be a scheme. Let $f : X \to Y$ be a flat morphism of algebraic spaces over S. Then for $x \in |X|$ we have: x has codimension 0 in $X \Rightarrow f(x)$ has codimension 0 in Y.*

Proof. Via Properties of Spaces, Lemma 0BAQ and étale localization this translates into the case of a morphism of schemes and generic points of irreducible components. Here the result follows as generalizations lift along flat morphisms of schemes, see Morphisms, Lemma 03HV. □

Date: stacks-project:spaces-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.